Perimeter and Area of Polygons on the Coordinate Plane

Directions

- **Perimeter** is the sum of the side lengths, and each side length comes from the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- **Area** can be found by the shoelace method: going once around the vertices in order, add every $x_i y_{i+1}$, subtract every $x_{i+1} y_i$, then halve the size of the result.
- Give exact values where they are whole numbers; otherwise round to two decimal places. Write the unit on every answer.
- Where a grid is provided, sketching the figure is optional — it is there so you can check that your answer is sensible.

1. Find the area of the triangle with vertices $A(0, 0)$, $B(9, 0)$ and $C(9, 12)$.

Answer: ________________ square units

2. A triangle has vertices $A(-3,2)$, $B(5,8)$ and $C(5,2)$.

(a) Find AB.

Answer: ________________ units

(b) Find BC.

Answer: ________________ units

(c) Find CA.

Answer: ________________ units

(d) Find the perimeter of the triangle.

Answer: ________________ units

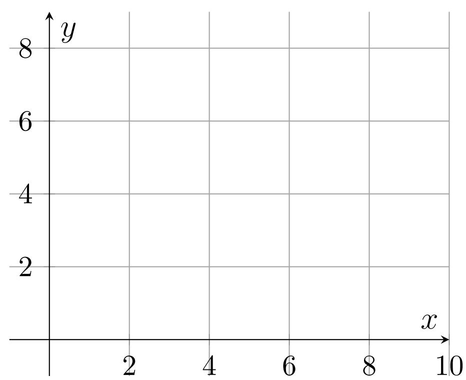
3. Find the area of the quadrilateral with vertices $(1,1)$, $(7,1)$, $(9,5)$ and $(3,5)$, taken in that order around the figure.

Answer: ________________ square units

4. A city park is mapped as a quadrilateral with corners at $(0,0)$, $(10,0)$, $(12,6)$ and $(2,8)$, taken in that order around the boundary. Each unit on the map is one metre. Find the area of the park, in square metres.

Answer: ________________ m^2

5. An L-shaped patio has corners at $(0,0)$, $(8,0)$, $(8,4)$, $(4,4)$, $(4,7)$ and $(0,7)$, taken in that order around the boundary. Each unit is one metre.



(a) Find the area of the patio, in square metres.

Answer: ________________ m^2

(b) Explain how you could check this answer by splitting the patio into two rectangles, and show that the two pieces add to your answer.

6. A quadrilateral has vertices $W(-2,1)$, $X(3,1)$, $Y(7,4)$ and $Z(2,4)$, taken in that order around the figure.

(a) Find WX.

Answer: ________________ units

(b) Find XY.

Answer: ________________ units

(c) All four sides of this figure are equal in length. Find its perimeter.

Answer: ________________ units

(d) Find the area of the figure.

Answer: ________________ square units

7. Two triangular plots are drawn on the same grid. The first has vertices $(0, 0)$, $(10, 0)$ and $(4, 6)$; the second has vertices $(0, 0)$, $(8, 0)$ and $(3, 9)$.

(a) Find the area of the first plot.

Answer: ________________ square units

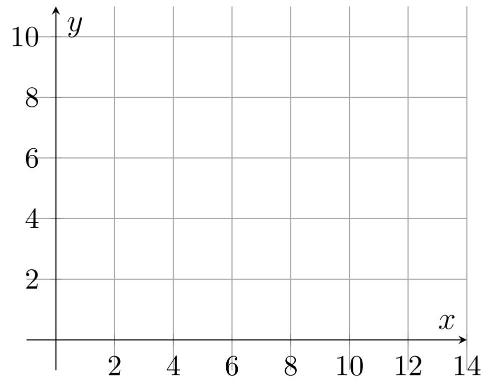
(b) Find the area of the second plot.

Answer: ________________ square units

(c) How much larger is the greater plot than the smaller one?

Answer: ________________ square units

8. A community garden is a pentagon with corners $A(0, 0)$, $B(12, 0)$, $C(12, 5)$, $D(6, 9)$ and $E(0, 5)$, taken in that order around the boundary. Each unit is one metre.



(a) Find the area of the garden, in square metres.

Answer: ________________ m^2

(b) Find the length of side $\overline{CD}$, in metres.

Answer: ________________ m

(c) Explain why the shoelace method gives the same area no matter which vertex you start from, provided you go around the boundary in order.